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Business Intelligence – Third Deliverable

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Deliverable 1

1.1 Introduction and Business Context

Pharma4All is a pharmaceutical company operating in a high-volume, data-driven environment. Currently, the company generates transactional data across products, pharmacies, and operators; however, it lacks a centralized mechanism to convert these raw records into actionable insights.

The goal of this project is to implement an end-to-end Self-Service Business Intelligence (BI) solution. By leveraging approximately three years of historical data, refreshed daily, this system will empower the administration to move from reactive record-keeping to proactive, data-backed decision-making. In this first phase, we establish the foundational Dimensional Model designed to support strategic, tactical, and operational analysis within the Microsoft Fabric ecosystem.

1.2 The Business Problem

The primary challenge at Pharma4All is fragmented information. While operational data exists in CSV format, it is not structured for analytical use. Decision-makers currently face several “blind spots”:

- Temporal Comparisons: Difficulty in performing Year-to-Date (YTD) or Year-Over-Year (YoY) performance analysis.
- Product Profiling: Inconsistent categorization makes it hard to distinguish the performance of Branded vs. Unbranded (Generic) products.
- Granularity vs. Aggregation: A lack of tools to seamlessly “drill down” from national sales totals to individual transaction-level details.
- Data Governance: The absence of Row-Level Security, which is critical for ensuring that regional managers only access data relevant to their specific locations.

1.3 Business Questions (BQs)

The following questions will be the main focus of our dimensional design and future BI reporting:

- BQ1 (Trends): How are sales (in terms of Amount and Quantity) evolving across months, quarters, and years?
- BQ2 (Performance): Which brands, laboratories, and products are the primary drivers of revenue?
- BQ3 (Comparative): How does current-year performance compare to the same period last year (in terms of both Amount and Quantity)?
- BQ4 (Product Mix): What is the sales volume (in terms of both Amount and Quantity) ratio between Branded and Generic products?
- BQ5 (Geographic): Which districts and municipalities represent the highest concentration of business volume?
- BQ6 (Operational): At a transaction level, which operators and Points of Suppliers (POS) are most active?

1.4 Source Data Audit & Discovery

Our discovery process involved auditing the source files and validating structure, completeness, and consistency. While the data is structurally sound, several “quality gates” must be addressed during the ETL phase.

Data Landscape Summary:

Source File	Record Count	Role	Key Findings
Sales (Pharmacy).csv	1 048 572	Fact	Contains one extreme outlier (~6M€) and one missing CNP.
Products.csv	72	Dimension	Missing header row; inconsistent “Generic” flags (Y, N, S); duplicate CNPs.
Locations.csv	133	Dimension	Missing header row; District field is 93% empty (requires lookup mapping).
Pharmacies (A & B)	23 (Total)	Dimension	Split across two files with Portuguese headers; requires merging and translation.
Operators.csv	15	Dimension	High data integrity; 4 missing emails; 5 operators with no sales records;
Point of Supply.csv	7	Dimension	Cleanest file in the dataset; ready for ingestion

Critical Observations for ETL:

1. Normalization: The Generic field in the Product file uses “S” (Sim), “Y”, and “N”. We will standardize this creating a binary variable *IsGeneric* = [True, False]. Additionally, we will add headers to the files that are missing that explanatory row (Products and Locations).
2. Geographic Enrichment: The District column is 93% missing in Locations.csv. A complete municipality-to-district lookup table (covering all 133 municipalities across Portugal's 18 continental districts) will be applied during ETL to fill all null values.
3. Integrity: The incomplete data for June 2023 must be flagged to prevent skewed averages in the final reports. The extreme transactional Amount outlier in FactSales will be removed, under the assumption that it is an error.

1.5 Data Modelling Methodology

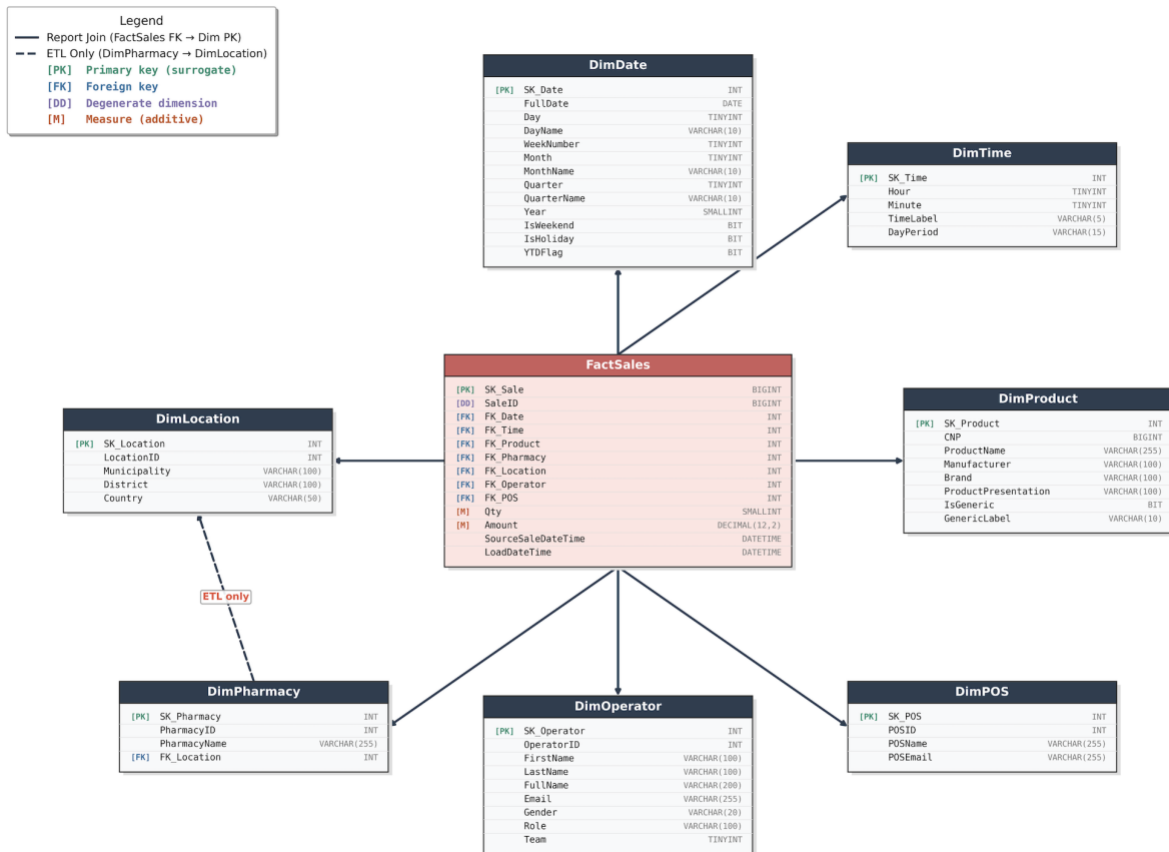
To design the dimensional model, Kimball dimensional modelling methodology was adopted, since it provides a business-oriented and reporting-friendly approach for structuring analytical data. This methodology is especially appropriate for the Pharma4All case because the project requires a star schema capable of supporting time-based analysis, product profiling, geographic aggregation, transaction-level drill-down, and secure reporting in Microsoft Fabric and Power BI.

Kimball's four-step process was applied as follows. First, the group selected the business process, defined as retail pharmaceutical sales transactions. Second, the grain of the model was declared as one row per individual sales line item, ensuring that the fact table preserves transactional detail while supporting aggregation at higher levels. Third, the relevant dimensions were identified to describe sales from multiple analytical perspectives, namely Date, Product, Location, Pharmacy, Operator, POS, and Time for intraday analysis. Finally, the central fact table, FactSales, was defined to store the quantitative business measures, namely Quantity and Amount, supported by foreign keys to the surrounding dimensions.

This methodology ensured that the final model remained aligned with the business questions while also satisfying dimensional modelling best practices. In particular, it allowed the group to design a star schema that is intuitive for end users, efficient for analytical querying, and suitable for future ETL and reporting stages.

1.6 Dimensional Model Design

The proposed model is a Star Schema - the gold standard for BI performance. This structure minimizes complex joins and maximizes the efficiency of DAX calculations.



Dimensions (The “Context”)

One row per individual sales transaction recorded at a pharmacy point-of-sale terminal. Each row represents the sale of one product unit-set at a specific pharmacy, processed by a specific operator, supplied through a specific point of supply, on a specific date and time.

We have defined seven dimensions, providing deep filtering capabilities:

- DimDate: Standard calendar attributes including Year, Quarter, and Month.
- DimTime: Hourly/Minute breakdown for intraday trend analysis.
- DimProduct: Contains the Product Hierarchy (*Manufacturer → Brand → Product*).
- DimOperator & DimPOS: Contextual data regarding the "Who" and "How" of the sale.
- DimLocation: Contains the Geographic Hierarchy (*Country → District → Municipality*).
- DimPharmacy: Details on specific retail outlets.

Fact Table (The “Metrics”)

- FactSales: The central hub containing foreign keys to all dimensions and the quantitative measures: *Quantity* and *Amount*.

- Degenerate Dimension: The *SaleID* is retained within the fact table to allow users to drill down to the specific receipt number without needing a massive “Transaction Dimension”.
- *CurrencyCode* identifies the currency in which the transaction amount was recorded. In the current Pharma4All dataset it is constant as EUR, so it is mainly retained as a technical/business traceability attribute rather than as an active analytical dimension.
- *SourceSaleDateTime* preserves the original transaction timestamp from the source system and is used to derive the analytical date and time keys. *LoadDateTime* is an ETL audit field that records when the row was loaded into the warehouse, supporting traceability, refresh validation, and future incremental-load control.

Key Design Decisions

To ensure long-term resilience and analytical flexibility, three structural decisions were made during modelling.

First, all dimension tables use auto-increment surrogate integer keys as primary keys, decoupling the warehouse from any future changes to source system identifiers.

Second, the transaction identifier (*SaleID*) is retained in FactSales as a degenerate dimension, a plain attribute rather than a foreign key, allowing analysts to drill through to individual receipts without requiring a separate, million-row transaction table. Additionally, we will create a surrogate key (*SK_Sale*, auto-incrementing at each transaction).

Third, to preserve a pure star schema in the reporting layer, the pharmacy-to-location relationship is resolved at ETL load time: each fact row receives a *SK_Location* derived from the source pharmacy record, enabling direct joins from FactSales to DimLocation without chaining through DimPharmacy.

Hierarchies & Depth

To support intuitive “Drill-Down” reporting, we have implemented:

1. Temporal: *Year* > *Quarter* > *Month* > *Day* (Depth: 4)
2. Product: *Manufacturer* > *Brand* > *Product* (Depth: 3)
3. Location: *Country* > *District* > *Municipality* (Depth: 3)

Relationships

All relationships from FactSales to a dimension table are Many-to-One: each fact row references exactly one dimension member, while each dimension member may appear in many fact rows.

Fact Column	Dimension	Type
<i>FactSales.FK_Date</i>	<i>DimDate.SK_Date</i>	<i>Many-to-One</i>
<i>FactSales.FK_Time</i>	<i>DimTime.SK_Time</i>	<i>Many-to-One</i>
<i>FactSales.FK_Product</i>	<i>DimProduct.SK_Product</i>	<i>Many-to-One</i>
<i>FactSales.FK_Pharmacy</i>	<i>DimPharmacy.SK_Pharmacy</i>	<i>Many-to-One</i>
<i>FactSales.FK_Location</i>	<i>DimLocation.SK_Location</i>	<i>Many-to-One</i>
<i>FactSales.FK_Operator</i>	<i>DimOperator.SK_Operator</i>	<i>Many-to-One</i>
<i>FactSales.FK_POS</i>	<i>DimPOS.SK_POS</i>	<i>Many-to-One</i>

The DimPharmacy–DimLocation link (used to resolve LocationKey during ETL) is not active in the reporting layer and does not create a snowflake join path for end users.

1.7 Description of Main Components

The dimensional model consists of one central fact table (FactSales) surrounded by seven dimension tables. Each component is described below in terms of its business purpose and analytical role.

FactSales: The Fact Table

FactSales is the analytical core of the model. It contains one row per individual sales transaction recorded across all 23 Pharma4All pharmacies, yielding a total of 1,048,572 rows covering approximately three years of history. The table stores two additive measures, Quantity (Qty) and Amount, that can be summed, averaged, and compared across any combination of dimensions without restriction,

DimDate: The Time Anchor

DimDate is the temporal backbone of the model. Every sales transaction is linked to a calendar date, enabling all time-based analysis. The table is fully pre-generated from the first date in the data through the end of the reporting period, ensuring continuity even for days with no sales activity. Two boolean flags add analytical depth: *IsHoliday* enables analysis of sales patterns on Portuguese public holidays, and *YTDFlag* provides a pre-computed basis for Year-to-Date accumulation.

DimTime: Intraday Granularity

DimTime captures the time-of-day dimension of each transaction. With 1,440 rows (one per minute), it is small but analytically valuable. The *DayPeriod* attribute (Morning / Afternoon / Evening / Night) groups transactions into operationally meaningful periods, enabling pharmacy managers to identify peak hours and optimise staffing and inventory replenishment schedules.

DimProduct: The Product Catalogue

DimProduct encodes the full pharmaceutical product catalogue. Its Product Hierarchy (Manufacturer → Brand → ProductName, depth 3) allows sales to be rolled up from individual products to brand and laboratory level. The *IsGeneric* attributes are central to BQ4 (branded vs generic profiling). Because the source data contained inconsistent coding (“S” for the Portuguese “sim” instead of “Y”), a standardisation step ensures all products are correctly classified before loading.

DimLocation: The Geographic Context

DimLocation maps each transaction to a Portuguese municipality and, through the Location Hierarchy (Country → District → Municipality, depth 3), enables geographic aggregation from national totals down to individual municipalities. The 93% missing District values in the source file are resolved during ETL via a complete municipality-to-district lookup table.

DimPharmacy: The Retail Outlet

DimPharmacy captures the characteristics of each of Pharma4All's 23 pharmacy outlets. *PharmacyName* is not a unique identifier, two distinct pharmacies share the name "Farmácia da Saúde" at different locations, so *PharmacyID* is always the authoritative identifier.

DimOperator: The Human Factor

DimOperator profiles the 15 individuals who process sales transactions across the network, enabling analysis of operator activity and productivity. The *Role* and *Team* attributes provide grouping capabilities for workforce performance analysis. Five operators (IDs 11–15) have no corresponding fact rows in the current dataset - they are retained as valid dimension members with zero-fact rows, ensuring future sales by these operators resolve correctly without any schema change.

DimPOS: The Distribution Channel

DimPOS identifies the seven Points of Supply through which pharmaceutical products are delivered to Pharma4All's pharmacies. It is a flat dimension with no hierarchy, but provides an important analytical lens: by filtering or grouping on DimPOS, analysts can compare sales volumes and revenue attributable to each distribution channel.

1.8 Conclusion & Next Steps

The dimensional model developed for Pharma4All provides a structured way of turning raw sales data into useful analytical information.

Through the combination of a central sales fact table and several business dimensions, the model supports a clear analysis of sales across time, products, locations, pharmacies and operational context. This creates the basis for more consistent exploration of business performance and for answering the key analytical questions identified in the report.

The next step is to implement this model in Microsoft Fabric in the already created Lakehouse and prepare the data to be loaded into the analytical structure. After that, the project can move forward to the reporting stage, where BI dashboards will be built on top of the model.

Deliverable 2

2.1 Integration with the Previous Delivery

The first delivery established the business context, the analytical requirements and the dimensional model for Pharma4All. This second delivery converts that conceptual model into an implemented Microsoft Fabric solution. The Lakehouse created in the first phase remains the landing zone for the source CSV files, while a Fabric Warehouse is used as the curated analytical layer. The ETL process transforms the raw files into a star-schema structure, loads the final dimension and fact tables, validates the load, and integrates the output of an advanced analytics notebook as an additional warehouse table.

The implementation also incorporates refinements identified after the first report. In particular, the warehouse keeps SaleID as a degenerate transaction identifier in FactSales and does not require a separate SaleKey. The analytical grain remains one row per individual sales line item, and the relationship structure is maintained through foreign keys from FactSales to the surrounding dimensions, keeping the model aligned with Kimball dimensional modelling principles while avoiding unnecessary keys in the fact table.

2.2 Fabric Architecture and ETL Scope

The solution was implemented in the same Fabric workspace used in the first delivery. The architecture separates raw storage, transformation logic, curated warehouse tables and advanced analytics output. This separation improves maintainability because each component has a clearly defined responsibility: the Lakehouse stores the source files, Dataflows Gen2 perform most deterministic transformations and warehouse loading, the Pipeline orchestrates execution order, and the Notebook produces the advanced analytics output.

Layer	Fabric artifact	Purpose
Raw layer	LH_Pharma4All / Files	Stores the original Pharma4All CSV files: Sales, Products, Locations, Pharmacies A and B, Operators and Point of Supply.
Transformation layer	Df_Load_Dimensions, Df_02_Load_FactSales	Cleans, standardizes, enriches and maps the source files into warehouse-ready tables.
Orchestration layer	Pipeline_Pharma4All	Runs the ETL components in the correct order and centralizes execution control.
Analytical storage layer	WH_Pharma4All / dwh schema	Stores the loaded star schema and the pharmacy segmentation output.
Advanced analytics layer	Python Notebook	Builds pharmacy-level behavioural features and applies K-Means clustering to create pharmacy segments.

Table 1 - Fabric architecture used in the second delivery

The implemented scope is a controlled full-refresh ETL process. Dimensions are loaded before the fact table so that the fact load can resolve foreign keys against already populated dimensions. The advanced analytics notebook runs after the core sales warehouse has been loaded, because the segmentation features depend on the transaction history and product attributes stored in the warehouse.

2.3 Presentation and Description of the Data Warehouse

The Warehouse implements the Pharma4All analytical model in the dwh schema. This avoids mixing curated analytical objects with default or staging objects and makes the final layer explicit for reporting. The implemented model follows a star-schema logic: FactSales is the central transactional table and the dimensions provide the analytical context needed for filtering, grouping and drill-down analysis.

Warehouse object	Type	Grain / key	Analytical role
DimDate	Dimension	One row per calendar date; key: SK_Date	Time intelligence, YTD and YoY analysis.
DimTime	Dimension	One row per minute; key: SK_Time	Intraday operational analysis by hour, minute and day period.
DimProduct	Dimension	One row per product/CNP; key: SK_Product	Product, brand, manufacturer/lab and branded/generic analysis.
DimLocation	Dimension	One row per municipality/location; key: SK_Location	Geographic aggregation and future RLS by location.
DimPharmacy	Dimension	One row per pharmacy; key: SK_Pharmacy	Pharmacy-level analysis and link to location.
DimOperator	Dimension	One row per operator; key: SK_Operator	Operational analysis by sales operator, role and team.
DimPOS	Dimension	One row per point of supply; key: SK_POS	Analysis by distribution / supply channel.
FactSales	Fact	One row per sales line item; FKs to all dimensions	Main quantitative table for Qty and Amount.
PharmacySegmentation	Advanced analytics output	One row per pharmacy; key: PharmacyID	Data-mining-based pharmacy segmentation for reporting.

Table 2 - Implemented Warehouse tables and analytical purpose

Key implementation columns include *FullDate*, *Month*, *Quarter*, *Year*, *IsHoliday* and *YTDFlag* in DimDate; *Hour*, *Minute* and *DayPeriod* in DimTime; *CNP*, *ProductName*, *Manufacturer*, *Brand*, *ProductPresentation* and *IsGeneric* in DimProduct; *LocationCode*, *Municipality*, *District* and *Country* in DimLocation; *PharmacyID* and *PharmacyName* in DimPharmacy; *OperatorID*, operator attributes, *Role*, *Team* and *Email* in DimOperator; *POSName* in DimPOS; and *PK_SaleID*, *FK_Date*, *FK_Time*, *FK_Product*, *FK_Pharmacy*, *FK_Location*, *FK_Operator*, *FK_POS*, *Qty*, *Amount*, *CurrencyCode*, *SourceSaleDateTime* and *LoadDateTime* in FactSales. The PharmacySegmentation table stores the notebook output, including *TotalSalesAmount*, *TotalQuantity*, *NumberOfTransactions*, *AvgTransactionAmount*, *BrandedSalesShare*, *ClusterID*, *SegmentLabel* and *ModelRunDate*.

FactSales preserves the transactional detail required by the original business questions. The two core measures are *Qty* (quantity) and *Amount*, both additive and therefore suitable for aggregation across date, product, location, pharmacy, operator and POS dimensions. *SaleID* is retained as a degenerate dimension so that users can drill through to the source receipt or transaction without creating a separate transaction dimension.

The dimensions are intentionally denormalized for analytical use. For example, DimProduct stores *Manufacturer*, *Brand* and *ProductName* in the same table, allowing the Product hierarchy to be used directly in Fabric. Likewise, DimLocation stores *Country*, *District* and *Municipality* together to support geographic drill-down and future Row-

Level Security filtering. This design remains consistent with the first delivery while now being physically implemented in the Warehouse.

2.4 ETL Setup: Pipelines, Dataflows and Notebook

The ETL setup uses a Pipeline as the orchestration artifact. The Pipeline coordinates the load sequence and prevents FactSales from being loaded before the dimensions it depends on. This order is essential because foreign keys in the fact table are resolved using the surrogate keys generated in the dimension tables.

Order	Pipeline activity	Purpose
1	SQL preparation	Prepares target tables for a controlled full refresh when needed.
2	Run Dataflow - Dimensions	Loads DimDate, DimTime, DimProduct, DimLocation, DimPharmacy, DimOperator and DimPOS.
3	Run Dataflow - FactSales	Loads the central sales fact after all dimension keys are available.
4	Run Notebook - Pharmacy Segmentation	Reads curated sales/product/pharmacy data and generates PharmacySegmentation.parquet.
5	SQL truncate dwh.PharmacySegmentation	Clears the previous segmentation result to avoid duplicates across model runs.
6	Copy Data activity	Copies Files/advanced_analytics/PharmacySegmentation.parquet into dwh.PharmacySegmentation.
7	Validation checks	Confirms row counts, non-null keys, no orphan foreign keys and successful completion.

Table 3 - ETL pipeline execution order

The Dataflows are responsible for deterministic ETL transformations. Splitting the process into a dimensions Dataflow and a fact Dataflow improves clarity and makes troubleshooting easier. It also reflects the dependency structure of the model: dimensions define the valid analytical members, while FactSales references those members through foreign keys.

The Notebook is not used to replace the standard ETL logic. Instead, it extends the Warehouse with an analytical output that cannot be produced only through basic cleansing operations. The resulting segmentation table is integrated back into the Warehouse, making the advanced analytics result available to the future semantic model and Power BI report.

2.5 Transformation Steps and Justification

The transformation layer was designed to solve the data quality issues identified during the first delivery and to prepare the source files for analytical querying. The following table summarizes the main transformations and their business justification.

Transformation	Implementation	Justification
Add / standardize headers	Products and Locations had missing header rows; columns were explicitly named before loading.	Prevents ambiguous fields and supports stable mapping into Warehouse tables.
Rename and translate columns	Source columns were renamed to a consistent English warehouse convention.	Improves readability and consistency across the model.

Append pharmacy files	Pharmacies A and Pharmacies B were combined into a single DimPharmacy source.	Creates one complete pharmacy dimension with all 23 pharmacy outlets.
Change data types	IDs, dates, times, quantities and amounts were converted to analytical types.	Avoids incorrect joins, sorting errors and numerical aggregation issues.
Clean CNP and remove duplicates	CNP was standardized and duplicated product business keys were removed.	Ensures each product resolves to one warehouse product key.
Replace values / conditional column	Generic values Y, N and S were standardized and product type was derived.	Supports reliable branded versus generic product analysis.
Geographic enrichment	Municipality-to-district mapping was applied through lookup/merge logic.	Enables geographic hierarchy and future RLS by location.
Date and time extraction	Sales Datetime was parsed into SourceSaleDateTime, FK_Date and FK_Time.	Supports both calendar and intraday analysis.
Surrogate key generation	Dimension keys were generated as SK_ fields while business keys were retained.	Decouples the warehouse from operational identifiers.
Fact-to-dimension merge	Sales rows were merged with dimensions to retrieve all FactSales foreign keys.	Guarantees the fact table joins directly to each analytical dimension.
Invalid record filtering	Invalid or non-resolvable rows and predefined extreme values were filtered according to quality rules.	Prevents distorted sales KPIs and orphan fact rows.
Add load metadata	LoadDateTime and ModelRunDate fields were added where appropriate.	Allows refresh tracking and reproducibility checks.
Remove extreme transaction	Transaction with extreme outlier amount identified in the first delivery was removed.	As state before, we assume that it was a mistake.

Table 4 - Main ETL transformations and justification

The implemented transformations cover a varied set of techniques required in the delivery: append operations, merges/lookups, replace values, conditional columns, group-by logic, duplicate removal, type conversion, filtering and derived columns. These steps are not only technical corrections; they directly support the business questions: for example, standardizing the generic flag is required for branded versus generic profiling, mapping districts is required for geographic analysis, and extracting date/time keys is required for monthly, quarterly, yearly and intraday reporting.

2.6 Advanced Analytics Notebook: Pharmacy Segmentation

The advanced analytics component is a Python notebook that performs pharmacy segmentation using transaction-enriched behavioural features. The main libraries are pandas and numpy for data manipulation, and scikit-learn for scaling and clustering.

The objective is to create a new analytical table that classifies pharmacies according to their sales behaviour. This output supports future reporting by allowing managers to compare high-value, standard and low-value pharmacy segments and to analyze whether different segments show different product mixes, transaction intensities or branded/generic preferences.

Step	Notebook operation	Purpose
1	Load curated sales/product/pharmacy data	Use the already transformed warehouse/lakehouse data as the input layer.
2	Aggregate by PharmacyID	Create one feature row per pharmacy.
3	Engineer behavioural features	Compute total amount, total quantity, number of transactions, average transaction amount, product variety, monthly sales intensity, branded/generic share and recency.
4	Scale numerical variables	Standardize features before clustering so that high-magnitude sales variables do not dominate the distance metric.
5	Apply K-Means	Assign each pharmacy to a cluster based on similar sales behaviour.
6	Label clusters	Translate numerical clusters into interpretable business labels, such as High-Value, Standard and Low-Value.
7	Persist output	Save PharmacySegmentation.parquet under Files/advanced_analytics and load it into dwh.PharmacySegmentation.

Table 5 - Pharmacy segmentation notebook workflow

Output column	Meaning
PharmacyID	Business key used to connect the segment result to DimPharmacy.
TotalSalesAmount	Total sales generated by the pharmacy over the historical period.
TotalQuantity	Total quantity sold by the pharmacy.
NumberOfTransactions	Count of sales transactions associated with the pharmacy.
AvgTransactionAmount	Average monetary value per transaction.
ProductVariety	Number of distinct products sold by the pharmacy.
MonthlySalesIntensity	Average sales intensity normalized by active months.
BrandedSalesShare	Share of sales associated with branded products.
GenericSalesShare	Share of sales associated with generic/unbranded products.
RecencyDays	Days since the pharmacy last recorded a transaction.
ClusterID	Numerical K-Means cluster identifier.
SegmentLabel	Business-friendly interpretation of the cluster.
ModelRunDate	Timestamp identifying when the segmentation was generated.

Table 6 - Columns produced by the advanced analytics notebook

2.7 Validation of the Loading Process

Validation is required to confirm that the ETL process did not only run successfully from a technical point of view, but also produced analytically consistent warehouse tables. The validation checks focus on row counts, key integrity, duplicate detection, measure plausibility and segmentation integration.

Validation area	Check	Reason
Row-count reconciliation	Compare source record counts with loaded warehouse records after documented exclusions.	Confirms that the full load is complete.
Dimension uniqueness	Check that each business key appears once in each dimension after deduplication.	Prevents many-to-many ambiguity in reporting.
Foreign key completeness	Check that FK_Date, FK_Time, FK_Product, FK_Pharmacy, FK_Location, FK_Operator and FK_POS are not null in valid fact rows.	Prevents orphan fact records.
Measure plausibility	Check that Qty and Amount are numeric and that invalid extreme values are excluded according to ETL rules.	Prevents distorted sales KPIs.
Date coverage	Verify that DimDate covers the complete source sales period.	Supports continuous time-series reporting.
Segmentation completeness	Check that each pharmacy receives at most one current segment row in dwh.PharmacySegmentation.	Guarantees correct use in Power BI slicers and visuals.
Pipeline execution status	Confirm that each activity finished successfully in the Fabric Pipeline monitor.	Provides operational evidence of ETL success.

Table 7 - ETL validation checks

2.8 Readiness for the Reporting Delivery

The second delivery creates the technical base required for the final reporting stage. The Warehouse now exposes a clean star schema that can be used to build the semantic model. The relationships from FactSales to the dimensions support the original business questions, while the PharmacySegmentation table enables a visual integration of advanced analytics results in the final Power BI report.

Business need	Warehouse objects	Reporting use
BQ1 - Sales trends	DimDate + FactSales	Monthly, quarterly, yearly and YTD sales amount and quantity.
BQ2 - Product performance	DimProduct + FactSales	Sales by manufacturer/lab, brand and product.
BQ3 - Current year vs previous year	DimDate + FactSales	Time intelligence measures in the semantic model.
BQ4 - Branded vs generic mix	DimProduct + FactSales	Sales amount and quantity split by GenericFlag/ProductType.
BQ5 - Geographic concentration	DimLocation + FactSales	Sales by country, district and municipality.

BQ6 - Operational activity	DimOperator, DimPOS + FactSales	Transactions, quantity and amount by operator and point of supply.
Advanced analytics visual	PharmacySegmentation + DimPharmacy + FactSales	Segment-level comparison of sales behaviour and product mix.

Table 8 - Link between Warehouse objects and final reporting needs

For the third delivery, the semantic model will mark DimDate as the date table, create the required DAX measures, hide technical key columns from end users, configure hierarchies for Date, Product and Location, and define the Row-Level Security role over DimLocation. The PharmacySegmentation table will be included in at least one report visual so that the advanced analytics output is visible and connected to the original business problem.

2.9 Extra: Incremental Load

As an extra ETL development, an extra pipeline includes an incremental load mechanism for FactSales. Instead of reloading the full sales table every time, the process only loads new sales records that have not yet been inserted into the Warehouse.

This mechanism is controlled by the etl.Watermark table. For the Load_FactSales process, the table stores the last successfully loaded sale identifier in the column *LastSaleID*. The pipeline then compares this value with the *SaleID* from the incremental staging table and only inserts records with a higher value.

The incremental process follows this logic:

1. Clear the previous incremental staging batch;
2. Load only new sales into stg.FactSales_IncrementalBatch.
3. Insert only non-existing records into dwh.FactSales.
4. Update etl.Watermark with the latest loaded *PK_SaleID*.

After the final pipeline run, the validation showed that dwh.FactSales contained 1,048,572 rows, with $\text{MAX}(\text{PK_SaleID}) = 1,048,572$. The etl.Watermark table was also updated to *LastSaleID* = 1,048,572, confirming that the incremental control value matches the latest loaded transaction.

This confirms that the incremental load is working correctly. If the pipeline runs again and no new sales exist, the staging table remains empty and no new rows are inserted. This makes the ETL process more efficient and closer to a realistic production scenario, where only newly arrived transactional data should be loaded.

2.10 Conclusion

This second delivery implements the Pharma4All dimensional model in Microsoft Fabric and moves the project from design to execution. The raw files are stored in the Lakehouse, transformed through Dataflows Gen2, loaded into a curated Warehouse and orchestrated through a Pipeline. The Warehouse contains the main star-schema tables required for sales analysis and an additional dwh.PharmacySegmentation table generated by an advanced analytics notebook.

The implemented ETL process addresses the main data quality issues identified in the first delivery: missing headers, inconsistent generic flags, split pharmacy files, missing district values, duplicate product keys, date/time parsing and fact-to-dimension key resolution. The validation strategy ensures that the Warehouse is not only populated, but also reliable for analytical use.

The solution is now ready to support the final delivery. The next step is to build the semantic model and Power BI reports on top of the Warehouse, using the implemented dimensions, measures and segmentation output to answer the business questions defined in the first report.

Deliverable 3

3.1 Integration with the Previous Deliveries

The first delivery established the analytical foundation by defining six core Business Questions and designing a Kimball-compliant star schema, structured around a central FactSales table and seven surrounding dimensions.

The second delivery implemented this model in Microsoft Fabric: source files were transformed through Dataflows Gen2, loaded into the Warehouse (WH_Pharma4All) via Pipeline_Pharma4All, and extended with a PharmacySegmentation table generated by a K-Means clustering notebook.

This third and final delivery builds directly on top of that foundation. The curated star schema tables serve as the data source for a Semantic Model built in Microsoft Fabric, which configures the analytical layer for reporting, defining hierarchies, DAX measures, and data formatting. On top of the Semantic Model, a Power BI report was developed across four analytical pages, each targeting specific Business Questions through interactive visualizations, plus one homepage that connects all others. The PharmacySegmentation output from the notebook is integrated directly into the reporting layer, connecting the data mining results to a dedicated report page. Together, these components complete the end-to-end Self-Service BI solution for Pharma4All.

3.2 Semantic Model & DAX Measures

Semantic Model Configuration

The Semantic Model, which can be seen in Figure 1, was built on top of the Fabric Warehouse using Direct Lake on SQL storage mode, connecting directly to the curated dwh schema tables without data import.

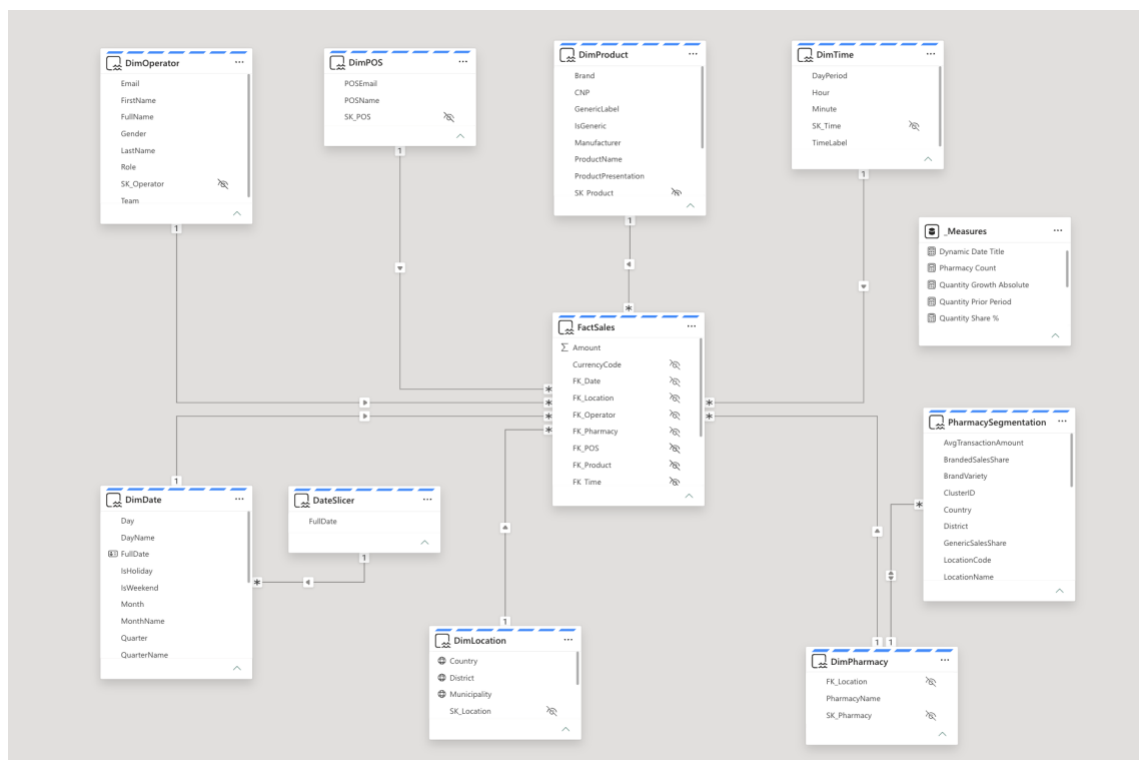


Figure 1 - Semantic Model (SM_Pharma4All)

The following configurations were applied:

- **Date Table:** DimDate was marked as the official date table, with FullDate set as the date column.
- **Hierarchies:** The previously identified hierarchies were configured:
 - Date (Year → QuarterName → MonthName → FullDate)
 - Product (Manufacturer → Brand → ProductName)
 - Location (Country → District → Municipality).
- **Hidden Columns:** All surrogate keys (SK_), foreign keys (FK_), and ETL metadata fields (such as *LoadDateTime* and *SourceSaleDateTime*) were explicitly hidden from the report view. This declutters the field pane, prevents end-users from accidentally aggregating identifier columns, and ensures that the administration only interacts with business-relevant attributes.
- **Data Types & Formatting:** For all columns the data types were checked and adjusted manually if it was not fitting by default. In some cases also the data category had to be changed, e.g. for the functionality of the location hierarchy.
- **Summarization:** Default summarization was set to Sum for *Qty* and *Amount*; non-additive columns such as names and labels were set to Don't Summarize. Most analytical aggregation, however, is handled directly through DAX measures in the _Measures table rather than relying on implicit column summarization.
- **_Measures Table:** Rather than using implicit measures directly from FactSales columns, a dedicated _Measures table was created to house all DAX measures explicitly. Base measures such as *Total Sales Amount* and *Total Quantity* act as the single entry point for all dependent measures, ensuring consistency, reusability, and easier maintenance across the model.

DAX Measures

All measures are stored in the _Measures table. The following table provides an overview of all measures, their logic, and their Purpose:

Measure	Logic	Purpose	BQs
Total Sales Amount	<i>SUM(FactSales[Amount])</i>	Base revenue measure; foundation for all dependent measures	BQ1, BQ2, BQ4, BQ5, BQ6
Total Quantity	<i>SUM(FactSales[Qty])</i>	Base volume measure; foundation for all dependent measures	BQ1, BQ4, BQ5, BQ6
Sales Prior Period	<i>CALCULATE([Total Sales Amount], SAMEPERIODLASTYEAR(DimDate[FullDate]))</i>	Revenue for the equivalent period in the prior year	BQ3

Sales Growth Absolute	<i>[Total Sales Amount] - [Sales Prior Period]</i>	Absolute revenue difference vs. prior period	BQ3
Quantity Growth Absolute	<i>[Total Quantity] - [Quantity Prior Period]</i>	Absolute volume difference vs. prior period	BQ3
Quantity Prior Period	<i>CALCULATE([Total Quantity], SAMEPERIODLASTYEAR(DimDate[FullDate]))</i>	Volume for the equivalent period in the prior year	BQ3
Pharmacy Count	<i>DISTINCTCOUNT(DimPharmacy[SK_Pharmacy])</i>	Number of pharmacies	BQ5, BQ6
Dynamic Date Title	<i>VAR StartDate = FORMAT(MIN('DimDate'[FullDate]), "MM/yyyy") VAR EndDate = FORMAT(MAX('DimDate'[FullDate]), "MM/yyyy") RETURN "Total Sales Amount (" & StartDate & " - " & EndDate & ")"</i>	Generates a dynamic visual title reflecting the selected date range	-
Sales Amount Share %	<i>DateSlicer</i> pattern - see below	Revenue share of a dimension member within the selected date range	BQ1, BQ2, BQ4
Quantity Share %	<i>DateSlicer</i> pattern - see below	Volume share of a dimension member within the selected date range	BQ1, BQ4

Table 1 - Description of Measures

Two of the more advanced measures in the semantic model are *Sales Amount Share %* and *Quantity Share %*. Both compute the percentage contribution of the current selection relative to the total within the user-selected date range.

A key design decision was to introduce a dedicated *DateSlider* table, connected to *DimDate* via a single-direction relationship. The date slicer in the report is bound to *DateSlider* rather than directly to *DimDate*. This separation allows the measures to read the selected date boundaries explicitly and reapply them to *DimDate* after removing all other active filters ensuring the denominator always reflects the full total for the chosen period, regardless of what else is selected on the page. Note that we decided to filter out the full year of 2020.

The practical result is a two-level selection pattern: the date range slicer defines the baseline (100%), while clicking any visual element, such as a bar in the chart, narrows the numerator without affecting the denominator. As shown in Figure 2, selecting August within the dataset range of 2022 correctly returns a share of 8.5%, reflecting that month's contribution to the whole year.

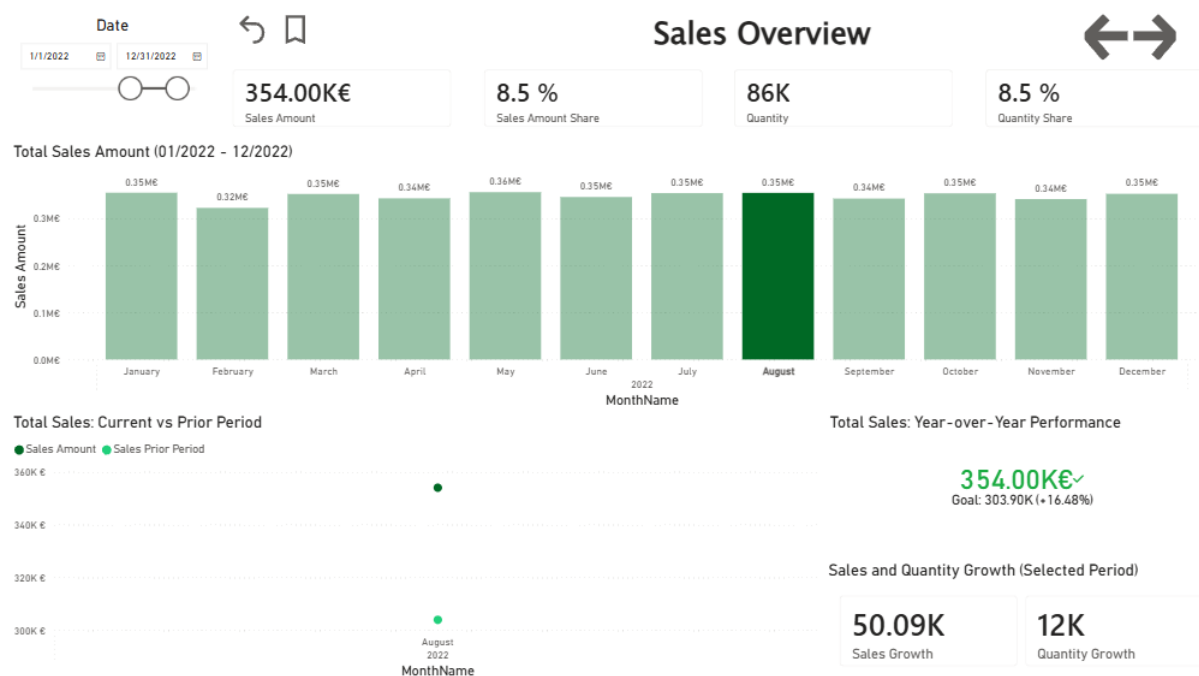


Figure 2 - Page 1: DateSlider Demonstration

Both measures are structured with the same logic:

Sales Amount Share % =

VAR _MinDate = *MIN(DateSlider[FullDate])*

VAR _MaxDate = *MAX(DateSlider[FullDate])*

VAR _Numerator = *[Total Sales Amount]*

VAR _Denominator =

CALCULATE (

[Total Sales Amount],

REMOVEFILTERS() ,

DimDate[FullDate] >= _MinDate &&

```

DimDate[FullDate] <= _MaxDate

)

RETURN

DIVIDE(_Numerator, _Denominator)

```

_MinDate and *_MaxDate* capture the boundaries of the selected date range from the *DateSlicer* table. *_Numerator* holds the sales total under the current filter context, while *_Denominator* recomputes the total with all filters removed but the date range explicitly restored. *DIVIDE* then returns the ratio, handling any division-by-zero cases gracefully.

3.3 Dashboard Architecture and Design

The Power BI report is structured into four distinct analytical pages. Rather than functioning as static reports, these pages are designed to address the project's core Business Questions (BQs) through interactive data discovery.

The following sections detail the contents of each page, how to interpret the resulting visualizations, and how the end-user can interact with the data to extract strategic insights.

Page 1: Sales Overview

Business Focus: BQ1 (Trends) and BQ3 (Comparative).

Contents: Serving as the executive summary of the report, this page displays macro-level KPIs for Total Sales Amount, Sales Amount Share, Total Quantity, and Quantity Share. It features a chronological bar chart tracking monthly sales performance across the full date range, and a dual-series area chart overlaying current-period sales against the prior period. A KPI gauge visual tracks total sales of the current period against the last year, and two card visuals report absolute Sales Growth and Quantity Growth for the selected period.

Interpretation: The primary goal is to assess the overall commercial performance of Pharma4All. The monthly bar chart addresses BQ1 by enabling the identification of seasonal patterns, growth trajectories, and anomalies across months and years. The area chart addresses BQ3 by making deviations between the current and prior period immediately visible at a glance. The KPI gauge contextualizes total sales, while the Growth cards quantify the absolute delta in both Amount and Quantity for any selected period.

Interaction: Users can utilize the date slider to isolate specific quarters or years, or click directly on a single bar in the monthly chart to filter all visuals to that specific month. Both interactions dynamically recalculate all KPIs and growth indicators, allowing decision-makers to pinpoint exact periods of underperformance or exceptional growth.

Page 2: Product Analysis

Business Focus: BQ2 (Performance) and BQ4 (Product Mix).

Contents: This page analyzes the full product portfolio using three complementary visuals. A Decomposition Tree traverses the product hierarchy (Manufacturer → Brand → Product), displaying Sales Amount at each level. A clustered bar chart shows absolute Quantity and Sales Amount split by GenericLabel (Generic vs. Branded), and a donut chart displays the relative Sales Amount share between the two categories.

Interpretation: The Decomposition Tree addresses BQ2 by making the revenue contribution at each level of the product hierarchy directly visible allowing users to identify which manufacturers, brands, and individual products are the primary drivers of sales, and how concentrated or distributed that revenue is across the portfolio. The clustered bar chart and donut chart jointly address BQ4 by presenting both the absolute volumes and relative proportions of Branded versus Generic products in terms of both Amount and Quantity, giving a complete picture of the product mix.

Interaction: The Decomposition Tree is highly interactive; users can expand or collapse individual nodes to navigate from the total sales figure down to specific brands and products. Selecting a segment in the donut chart cross-filters the entire page, restricting all visuals to either the Generic or Branded segment, which allows for a focused comparison of performance within each category.

Page 3: Geographic & Operational

Business Focus: BQ5 (Geographic) and BQ6 (Operational).

Contents: This page combines spatial and operational analysis through four visuals. A horizontal bar chart ranks districts and municipalities by total Sales Amount. An Azure Maps visual plots sales density across Portuguese territory using proportionally sized bubbles. A donut chart breaks down total Sales Amount by Point of Supply (POS). A pharmacy-level table lists each pharmacy with its individual total Sales Amount and total Quantity. A dynamic text box provides a natural-language summary of the totals for the current selection.

Interpretation: The bar chart addresses BQ5 by providing a ranked view of which districts and municipalities concentrate the highest business volume. The map reinforces this spatially, making regional clusters and underrepresented areas immediately visible. Together they allow management to assess geographic market penetration at both the district and municipality level. The donut chart addresses BQ6 by showing the relative revenue contribution of each POS channel, revealing the degree of operational dependency on individual distributors. The pharmacy table extends BQ6 to the individual outlet level, enabling granular performance comparisons across all pharmacies.

Interaction: Users can click on a district in the bar chart or drill down to municipality level, which dynamically causes the map to highlight the selected region and cross-filters both the POS donut chart and the pharmacy table accordingly. The filtering also works in reverse: selecting a POS segment in the donut chart filters the map and table to that specific distribution channel. Additionally, hovering over any bubble on the map triggers a custom Report Page Tooltip, which displays a localized sales breakdown of *Generic* vs *Branded* products without requiring the user to navigate away from the page.

Page 4: Pharmacy Segmentation

Business Focus: This page goes beyond the core business questions and integrates the data mining output developed in the second delivery directly into the reporting layer.

Contents: This page visualizes the output of the K-Means clustering algorithm applied to the pharmacy features. A PCA scatter plot positions each pharmacy in a two-dimensional reduced feature space, color-coded by segment label (*High-value*, *Standard*, *Low-value*). A clustered bar chart with a secondary axis compares the average Monthly Sales Intensity and average Transaction Amount across the three segments. A pharmacy-level table lists each pharmacy with its assigned segment label, total Quantity, and total Sales Amount. Two KPI cards display the average Branded Sales Share and the total Pharmacy Count for the current selection.

Interpretation: The K-Means algorithm was applied to behavioral features, including total sales, transaction count, average transaction amount, product variety, monthly sales intensity, and

branded/generic share. PCA was then used to reduce this high-dimensional feature space to two principal components for visualization, allowing the cluster separation to be assessed visually. This allows management to move beyond total revenue and understand how pharmacies operate behaviorally revealing the strategic differences between pharmacies. For instance, it identifies whether a segment achieves its status through sheer transaction volume or through premium, high-ticket sales. The KPI cards further enrich this by surfacing the branded product preference within the selected segment.

Interaction: To facilitate seamless strategic presentations, this page utilizes Power BI's Bookmarks and Selection pane functionality. End-users can trigger predefined bookmarks to instantly isolate a specific cluster, automatically hiding secondary visuals to focus the narrative entirely on the most critical business segment. Additionally, clicking on a data point in the scatter plot or a row in the table cross-filters all other visuals on the page to that pharmacy's segment.

3.3.1 Executive UI/UX & Interactive Discovery

To elevate the dashboard from a standard reporting tool to an executive-level application, several advanced User Interface (UI) and User Experience (UX) strategies were implemented.

First, a professional custom corporate color palette was established globally across the report. This ensures high-contrast readability, such as distinctly separating the "High-Value" from "Standard" clusters in the PCA plot, and maintains a cohesive, institutional aesthetic rather than relying on default software colors.

Additionally, to create an app-like navigation experience, an "Executive Portal" landing page was created using centralized Page Navigator buttons. Furthermore, Next/Previous arrow buttons were placed on every report page, allowing stakeholders to seamlessly page through the data story without using the native Power BI page tabs.

Finally, advanced Power BI Bookmarks and Selection pane logic were utilized to create Insight Highlights that capture specific strategic narratives. Users can trigger predefined bookmarks to take a deeper look at the variance from the first quarter of 2022, focus on the Generic product volume engine, highlight the specific geographic distribution of the Braga municipality, or isolate the "High-Value" pharmacy cluster. A universal "Reset View" mechanism was also built into every page to ensure users can safely clear all cross-filters and return to the macro view.

3.4 Analysis and Discussion of Outputs

The interactive BI solution successfully addresses the six core Business Questions (BQs) established at the project's inception. Based on the ~9.8M€ in total Sales Amount and ~2.38M in total Quantity processed over the historical period, the following strategic insights were extracted using the report's interactive discovery features:

Macro Trends & Comparative Growth (BQ1 & BQ3): The Sales Overview reveals a consistent baseline of monthly revenue, generally fluctuating between 300K€ and 400K€. To assess macro-level health against historical baselines, the "**Q1 2022 Variance**" bookmark can be triggered to isolate specific comparative periods.

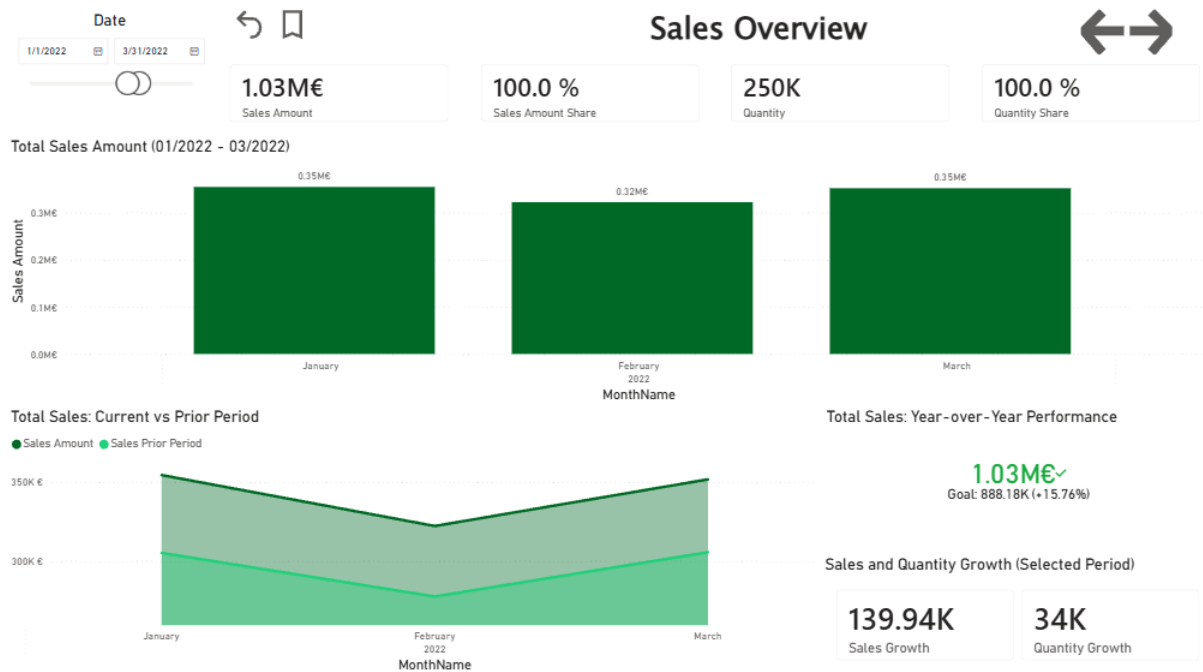


Figure 3 - Page 1: Q1 2022 Variance

As highlighted in this bookmarked view, the Year-over-Year (YoY) performance metrics indicate strong growth trajectories. During the selected timeframe, total sales reached 1.03M€ against a prior-period baseline of 888.18K€, representing a robust 19.33% growth rate.

Product Performance & Market Mix (BQ2 & BQ4): Analysis via the Decomposition Tree highlights that laboratories such as Sanofi, Roche, and Novo Nordisk are the primary drivers of macro revenue. There is a clear strategic divide in the portfolio: Branded products dominate the total revenue share, accounting for 54.5% of the total Sales Amount. However, activating the "**Generic Focus**" bookmark filters the hierarchy to reveal a different operational reality.

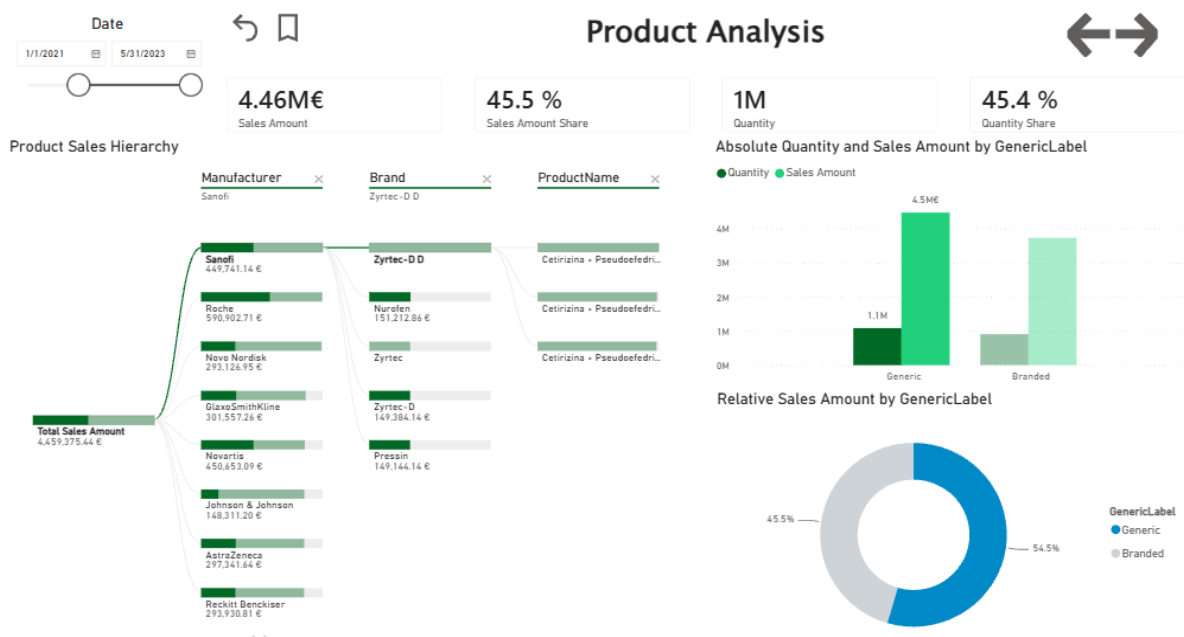


Figure 4 - Page 2: Generic product volume engine

This view proves that while branded pharmaceuticals are critical for maintaining high profit margins, generics are the true engine driving physical transaction volume and daily foot traffic.

Geographic Concentration & Operational Reliance (BQ5 & BQ6): When looking at the map and municipality rankings, it becomes clear that sales are not spread evenly across the country. Instead, the vast majority of revenue comes from a few leading markets. Beja is the strongest performing municipality by a wide margin, generating roughly 0.87M€ in sales, followed by key northern hubs like Braga (0.44M€). To understand how our supply chain supports these high-demand areas, the **"Braga Details"** bookmark can be triggered to cross-filter the Point of Supply (POS) breakdown specifically for this region.

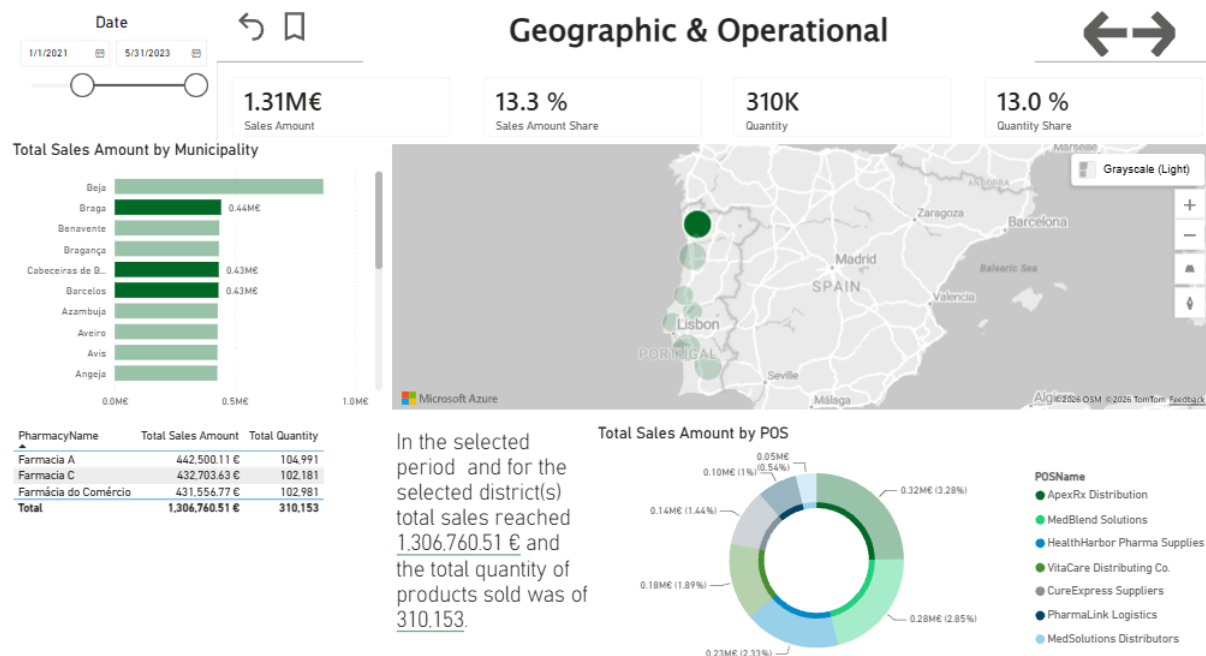


Figure 5 - Page 3: Geographic distribution of the Braga municipality

This filtered view isolates 1.31M€ in regional sales and demonstrates the localized operational reliance on top-tier distributors to keep this critical northern market stocked. Within this specific regional network, the top distributors mirror our national supply chain: the majority of the regional distribution is handled by ApexRx Distribution (0.32M€) and MedBlend Solutions (0.28M€). This proves that our macro-level supply leaders are successfully dominating the localized distribution in our most profitable geographic hubs.

Advanced Analytics: Pharmacy Behavior: Integrating the K-Means clustering output provided the most critical strategic insight of the project. By triggering the **"High-Value Pharmacy"** bookmark, the PCA scatter plot and associated visuals isolate the top-performing pharmacy cluster, which currently consists of a single flagship location.

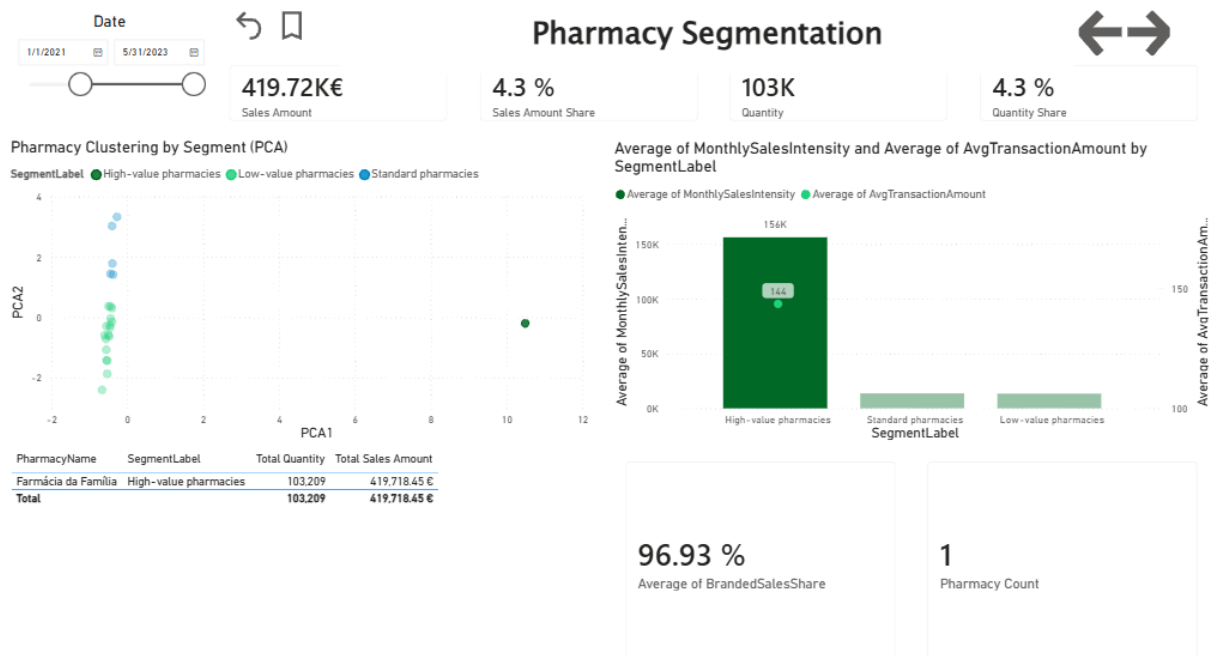


Figure 6 - Page 4: High-Value pharmacy cluster

This specific view proves that High-Value pharmacies operate on a fundamentally different business model. As a whole, they average an exceptional 156K€ in Monthly Sales Intensity and a 144€ Average Transaction Amount, compared to just 14K€ and 13€ for Standard and Low-value pharmacies. Crucially, when we isolate **Farmácia da Família**, the KPI cards reveal a staggering **96.93%** Branded Sales Share. This proves that extreme branded product penetration is the primary catalyst for our absolute highest-tier retail performance.

3.5 Conclusion

This project delivered a complete, end-to-end Self-Service Business Intelligence solution for Pharma4All across three sequential deliveries, each building directly on the output of the previous one.

The first delivery established the analytical foundation: six Business Questions were defined to capture the company's core informational needs, and a Kimball-compliant star schema was designed around a central FactSales table and seven surrounding dimensions. Key design decisions, such as the use of surrogate keys, the degenerate *SaleID* dimension, and the denormalized geographic and product hierarchies, were made deliberately to support both reporting performance and long-term maintainability.

The second delivery implemented this model in Microsoft Fabric. Raw CSV source files were ingested into a Lakehouse, transformed through Dataflows Gen2 to resolve the data quality issues identified during the first delivery, and loaded into the curated WH_Pharma4All warehouse via Pipeline_Pharma4All. A Python notebook extended the solution beyond standard ETL by applying K-Means clustering to engineer behavioral pharmacy segments, persisting the result as an additional analytical table. An incremental load mechanism was also implemented as an extra development, bringing the pipeline closer to a production-grade scenario.

The third and final delivery completed the reporting layer. A Semantic Model was configured in Direct Lake on SQL mode, with hierarchies, hidden technical columns, data type formatting, and a dedicated _Measures table housing all DAX logic. A set of measures covering time intelligence, growth comparison, product mix analysis, and share percentages was developed to answer each business question analytically. The Power BI report spans five pages in total (Home Page, Sales Overview, Product Analysis, Geographic & Operational, and Pharmacy Segmentation) each combining multiple visual types with cross-filtering and drill-down interactions. The data

mining output from the second delivery is directly integrated into the final reporting page, closing the loop between advanced analytics and business interpretation.

Across all three deliveries, the project demonstrated how a structured BI methodology, combined with the capabilities of Microsoft Fabric and Power BI, can transform fragmented operational data into a coherent, interactive analytical system that supports strategic, tactical, and operational decision-making.

References

[1] Kimball Group, Dimensional Modeling Techniques - Four-Step Dimensional Design Process.

[2] Course materials T1, T2, T3 and T4 - Business Intelligence, Data Warehousing, Dimensional Modelling and ETL.